



Research Paper

Layered-averaging systems and linear reward emergence in nonlinear financial markets: A structural interpretation

Li-Yung Chen

Department of Surgery, Gangshan Branch of Zuoying Armed Forces General Hospital, Kaohsiung 820, Taiwan

ARTICLE INFO

Keywords:

Layered averaging
Linear reward emergence
Structural interpretation
Nonlinear market dynamics
Depth-state transitions

ABSTRACT

Financial markets exhibit strongly nonlinear and non-stationary dynamics. However, in long-horizon live execution on spot digital-asset markets under bounded operational constraints, layered-averaging systems (often called “martingale” in trading practice, but distinct from the probabilistic martingale in measure theory) may produce a stable near-linear accumulation of realized reward. In this study, $Reward(t)$ denotes the cumulative realized profit from completed execution cycles under fixed exit and position-sizing rules.

We provide an execution-level structural interpretation of this linear reward emergence. Viewing layered averaging as a rule-governed intervention that converts nonlinear price trajectories into cycle-wise realizations, we analyze real-capital execution logs across multiple spot asset pairs with several hundred completed cycles. Three empirical regularities recur: (i) realized price paths exhibit higher-order nonlinear curvature; (ii) cumulative exposure induced by layered entries grows convexly and is well approximated by a quadratic profile; and (iii) when execution remains within its controllable range, $Reward(t)$ concentrates around a stable first-order linear envelope across cycles, including shock-driven drawdowns.

To characterize departures from this envelope, we introduce a reward deviation term $R(t)$, an execution-derived accumulator of structural stress operationalized from logs (e.g., drawdown duration, depth persistence, and holding time). This yields the generalized representation:

$$Reward(t) = at + b - R(t),$$

where a denotes the systematic accumulation rate under fixed exit constraints, b captures intermittent reinforcement associated with deep-layer reversals during shock-driven cycles, and $R(t)$ encodes accumulated deviation from the linear reward envelope. Overall, the findings suggest that layered averaging reorganizes nonlinear market dynamics into approximately linear reward increments through structured intervention rather than prediction, providing a measurement-oriented explanation of a long-observed empirical behavior under nonlinear uncertainty.

1. Introduction

Financial markets evolve under strongly nonlinear and non-stationary dynamics. Over long horizons, price trajectories exhibit path dependence, structural breaks, and regime shifts that often violate stationary assumptions used in model-based design. These characteristics are especially pronounced in digital-asset spot markets, where volatility clustering and liquidity shocks shape the operational risk profile of long-horizon execution systems. Nonlinearity and multi-scale dependence in cryptocurrency markets are well documented; wavelet-based evidence shows time-varying co-movement structures and nonstationary drivers in Bitcoin dynamics [1]. Accordingly, we do not claim novelty in establishing market nonlinearity. Our contribution is an

execution-level geometric representation—via depth-indexed normalization and fan structures—that maps nonlinear drawdown curvature into state-dependent deformation and boundary behavior under bounded execution constraints.

Conventional downside-risk measures such as Value-at-Risk (VaR) and Expected Shortfall (ES) provide a common language for tail-loss description. Yet in crypto spot markets, discontinuous moves (e.g., 30–50%) and frequent regime shifts can destabilize any fixed “acceptable loss” threshold as an operational decision boundary, because dependence structures and volatility regimes shift rapidly. Prior evidence further suggests that tail-risk and diversification properties in crypto are time- and horizon-dependent rather than static [2]. Motivated by this instability, we define the boundary of interest in

E-mail address: chenlijon.tw@yahoo.com.tw.

<https://doi.org/10.1016/j.nexres.2026.101565>

Received 16 February 2026; Accepted 5 March 2026

Available online 6 March 2026

3050-4759/© 2026 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

operational terms—controllability—namely, the depth range within which an execution system remains rule-governed without discretionary intervention.

Drawdown-based measures also offer a useful baseline for describing adverse path behavior. In particular, conditional drawdown measures have been formulated to summarize tail drawdown severity and to incorporate drawdown constraints into optimization frameworks [3]. Our objective, however, is not to determine a universal tolerable-loss threshold or to solve a drawdown-constrained optimization problem. Instead, we examine how extreme downside realizations translate into execution-level stress, depth-state evolution, and eventual loss of controllability within a rule-based layered execution setting.

Layered-averaging systems—widely used in practice as multi-layer, DCA-style execution mechanisms—have historically been criticized as naive and risky, largely due to their association with gambling-style “martingale” narratives. In this paper, “martingale” is used strictly in the practice-based sense of layered averaging with predefined spacing and position sizing, and is explicitly distinguished from the probabilistic martingale concept in stochastic processes and from classical doubling schemes in gambling theory.

Despite this skepticism, long-horizon real-capital operation reveals a recurring empirical pattern: under bounded execution constraints, layered averaging can produce near-linear trajectories of $Reward(t)$, where $Reward(t)$ denotes the cumulative realized profit from completed execution cycles under fixed exit and position-sizing rules. Using execution logs from spot digital-asset trading across multiple asset pairs over more than 800 days and over 300 completed cycles, we document three recurring patterns: (i) realized price paths exhibit pronounced nonlinear curvature and regime shifts; (ii) cumulative exposure induced by layered entries grows convexly and is well approximated by a second-order profile under controlled execution; and (iii) when the system remains within its controllable range, $Reward(t)$ aligns with a stable first-order linear envelope at the cycle-aggregated level, including shock-driven drawdowns.

To interpret this behavior, we view layered averaging not as a probabilistic inevitability mechanism, but as a structured intervention that slices nonlinear drawdown trajectories into discrete execution cycles under fixed rules. Under predefined initiation conditions, the system selectively engages on the downside and converts drawdown–rebound segments into realized increments constrained by predetermined exits. Empirically, this concentration around a narrow linear manifold can be schematically expressed as:

$$Reward(t) \approx at + b,$$

where a is governed primarily by the exit specification, and b reflects intermittent reinforcement associated with deep-layer reversals during extreme displacement events.

To quantify departures from this idealized envelope, we introduce a deviation term $R(t)$, an execution-level proxy for accumulated structural stress. Rather than serving as a forecasting device, $R(t)$ measures deviations from the linear reward envelope arising from prolonged drawdown duration, depth persistence, extended holding time, and parameter–market misalignment. This yields the generalized representation:

$$Reward(t) = at + b - R(t).$$

Within this framing, sustained stress regimes increase $R(t)$, while post-capitulation reversals may reduce effective stress by enabling recovery at deeper entry levels. Whereas much of the literature approaches nonlinear dynamics and risk boundaries through forecasting, optimization, or learning-based methods, the present study focuses on how near-linear reward patterns empirically emerge at the execution level without relying on prediction, under bounded operational constraints.

Overall, this study does not propose a new trading algorithm or claim

universal profitability. Instead, it offers a measurement-oriented structural interpretation of linear reward emergence in spot digital-asset markets under bounded execution constraints. The subsequent sections formalize the framework, define operational proxies for $R(t)$, and analyze cycle-level depth-state transitions to clarify how reward regularity arises under nonlinear uncertainty.

2. Theoretical framework

2.1. Nonlinear market geometry (local descriptive view)

Financial price trajectories rarely evolve linearly over long horizons. In high-volatility digital-asset spot markets, volatility clustering, liquidity imbalance, and intermittent shocks can induce pronounced curvature and regime-dependent deformation [4–6]. Let $P(t)$ denote the market price at time t . Over a finite window, we use a local descriptive approximation to represent curvature:

$$P(t) \approx \sum_{m=0}^n c_m t^m, n \geq 3,$$

where higher-order terms summarize irregular, regime-dependent curvature. This polynomial form is not a generative price model; it is used only as a local proxy to visualize curvature over bounded intervals. This motivates the structural stance of this study: an execution system operating under bounded constraints necessarily interacts with price trajectories whose curvature is not adequately captured by linear or purely diffusive descriptions.

2.2. Layered averaging as a discrete accumulation mechanism

A layered-averaging execution system enters the market at a sequence of decreasing prices:

$$P_1, P_2, \dots, P_K, P_{K+1} < P_K.$$

Each layer is a discrete sampling point along the underlying nonlinear price path. Define cumulative deployed exposure at depth K as

$$E(K) = \sum_{k=1}^K w_k P_k,$$

where w_k denotes the effective position size (weight) at layer k . Here $E(\cdot)$ is cumulative deployed exposure as a function of depth and is not an expectation operator.

Under the configurations studied, spacing is fixed and entries are monotone by design; deeper layers typically carry larger weights. Consequently, $E(K)$ increases nonlinearly with depth. As shown empirically in Section 4.2, under controlled execution the exposure profile is well summarized by a quadratic form:

$$E(K) \approx c_2 K^2 + c_1 K + c_0,$$

where c_2, c_1, c_0 are empirical coefficients estimated from execution logs and represent convexity, linear trend, and baseline exposure level under a fixed configuration.

Interpretation. Rather than claiming markets are fundamentally “unmodelable,” we adopt a pragmatic execution-level projection: global parametric validity over long horizons is unreliable, but local approximation and structural mapping remain feasible. Polynomial and quadratic forms are therefore used here as descriptive instruments to represent curvature and response under bounded execution, not as global market equations.

2.3. Execution-level reward mapping

We formalize the rule-governed mapping from realized price paths to cycle-level reward under fixed entry and exit constraints. Consider cycle

i , which activates K_i layers and exits at a predefined take-profit level $P_{\text{exit},i}$. The realized reward of a completed cycle is

$$\text{Reward}_i = \sum_{k=1}^{K_i} w_k (P_{\text{exit},i} - P_{k,i}),$$

where $P_{k,i}$ is the entry price of layer k in cycle i . Aggregating completed cycles over time yields $\text{Reward}(t)$, the cumulative realized profit generated under fixed entry, exit, and sizing rules.

Empirically (Section 4), $\text{Reward}(t)$ concentrates around a stable near-linear envelope:

$$\text{Reward}(t) \approx at + b.$$

This regularity does not rely on stationarity assumptions or noise averaging. It arises from execution structure. Two cycle-level mechanisms dominate:

(i) Shallow-depth dominance.

Most execution cycles terminate within the first one or two layers, producing rewards that lie within a relatively narrow range. These frequent shallow completions dominate the long-horizon accumulation rate and primarily determine the slope parameter a .

(ii) Deep-layer reinforcement.

Less frequent deep-depth cycles generate larger rebound-driven rewards due to increased accumulated exposure. Because such cycles typically exhibit longer holding times and elevated execution stress, their contribution is better interpreted as intermittent reinforcement reflected in the intercept b , rather than as a shift in the baseline accumulation rate a .

From an execution-systems perspective, layered averaging functions as a structured nonlinear response mapping that transforms higher-order drawdown curvature into approximately linear cycle-aggregated cumulative reward behavior, provided that execution remains within a controllable depth range.

2.4. Deviations from linearity: reward deviation $R(t)$

Real execution does not always adhere to the near-linear envelope. Prolonged downward drift, delayed rebounds, spacing misalignment, and adverse volatility regimes can produce flattening or downward bending relative to the envelope. To quantify these departures at the execution level, we introduce a non-negative deviation term $R(t) \geq 0$:

$$\text{Reward}(t) = at + b - R(t).$$

Here a denotes the baseline accumulation rate primarily governed by fixed exit constraints; b captures intermittent reinforcement associated with deep-layer reversals; and $R(t)$ encodes accumulated departures from the linear envelope attributable to execution-level structural stress.

Importantly, $R(t)$ is not proposed as a universal downside-risk metric (VaR/ES/EVaR) and not as a forecasting signal. It is an execution-derived deviation accumulator, anchored to log-constructible proxies that explain when and why near-linear reward accumulation degrades under bounded constraints.

A compact proxy set (main-text version). Let i index cycles. We define non-negative cycle-level stress components:

- Excess-depth: downside exceeds designed coverage

$$\begin{aligned} D_{\text{mkt},i} &= 1 - \frac{P_{\text{min},i}}{P_{0,i}}, D_{\text{max}} = 1 - (1 - x)^{L-1}, X_i \\ &= \max \{0, D_{\text{mkt},i} - D_{\text{max}}\}. \end{aligned}$$

- Time-under-stress: prolonged holding beyond reference

$$H_i = t_i^{\text{exit}} - t_i^{\text{start}}, U_i = \max \{0, H_i - H_0\}.$$

- Entry clustering (insufficient cooldown): accelerated layer consumption

$$F_i = \frac{1}{K_i - 1} \sum_{k=1}^{K_i-1} 1\{\Delta t_{i,k} < \tau\}.$$

- Inter-cycle overlap: insufficient recovery between cycles

$$\Delta \tau_i = t_{i+1}^{\text{start}} - t_i^{\text{exit}}, M_i = 1\{\Delta \tau_i < \tau_{\text{sep}}\}.$$

- Rebound deficit: insufficient rebound amplitude delaying exit

$$V_i = \max \{0, A_0 - A_i\}.$$

An interpretable uncalibrated aggregation is

$$R(t) = \sum_{i: t_i \leq t} (X_i + U_i + F_i + M_i + V_i + \dots),$$

where “...” denotes additional log-derived components if needed. The objective is auditability and interpretability rather than optimization: elevated $R(t)$ explains why the reward trajectory flattens or bends; when stress diminishes, deviation accumulation slows and near-linear behavior re-emerges. An operational proxy family for $R(t)$, computable from execution logs and intended purely as a diagnostic/audit interface, is provided in Appendix A.

2.5. Summary interpretation (execution-level system)

The framework can be summarized as:

- $P(t)$: nonlinear price trajectory (local curvature proxy)
- $E(K)$: convex (approximately quadratic) exposure accumulation with depth
- $\text{Reward}(t)$: near-linear reward envelope under bounded execution
- $R(t)$: deviation accumulator capturing execution-level structural stress

Together, these components describe layered averaging as a rule-governed input-output system that reorganizes nonlinear market dynamics into approximately linear realized reward trajectories within a controllable operating range. We refer to this execution-level framing as the Differential Martingale (DMT) interpretation.

3. Data and Experimental Design

This section describes the empirical dataset, system configuration, and analytical design used to instantiate the framework in Section 2. The unit of analysis is the execution cycle: a completed entry-to-exit sequence generated by fixed, rule-governed entry/exit logic. Encoding each cycle as an ordered set of depth states enables systematic analysis of (i) depth-dependent behavior, (ii) regime transitions, and (iii) structural deviations from the idealized linear reward envelope.

3.1. Dataset and market environment

The study evaluates an execution system as actually operated in live spot trading under bounded execution constraints. The objective is not to claim that layered averaging is universally stable or predictive, but to measure how execution-level controllability and reward-envelope

behavior emerge (or degrade) under real market paths.

A pre-deployment calibration phase was conducted more than two years before the analyzed live period. This phase acted as a survivability filter rather than an optimization procedure: candidate configurations were retained or removed based on operational breakdown under historically severe drawdowns (e.g., depth saturation, prolonged unresolved cycles), not on performance tuning. The live system analyzed here represents the surviving configuration after iterative elimination of structurally fragile settings. An illustrative example from this exploratory phase is provided in Fig. 1 to show the typical staircase-like reward accumulation and the sensitivity of realized depth to local price-path curvature.

The system was deployed in live spot trading from September 30, 2023 to December 10, 2025. During this period, 305 completed execution cycles were recorded using real capital. Although the deployment covered multiple spot asset pairs, the quantitative analyses in this manuscript focus on a single representative dataset (Asset-A/USDT), selected for (i) uninterrupted runtime and (ii) the largest number of completed cycles ($n = 305$). Cross-asset comparisons are intentionally not reported because the aim is execution-level measurement and controllability characterization within a coherent operating environment, rather than generalized performance claims.

All activity was conducted exclusively in spot markets without leverage or derivatives. This removes liquidation mechanics as a confound and ensures that observed dynamics arise from the interaction between realized price paths and the system's exposure accumulation.

Raw data include intraday price series, order-level transaction logs (timestamps, filled prices, filled sizes), position-depth records, and cycle-level realized reward. Timestamps were normalized and identifying details were anonymized.

3.2. Parameter selection and system configuration

The configuration used in the live dataset was retained through the survivability-filter process described above. The resulting system consists of (i) a rule-governed entry module, (ii) a bounded multi-layer averaging engine, and (iii) an integral-style trailing take-profit exit.

Entry logic. Entries are initiated by RSI-based micro-entry signals combined with an event-driven trailing-entry mechanism. The first layer is triggered when RSI falls below a predefined threshold. Subsequent

layers are not placed at fixed price intervals; instead, they are activated only when price reaches a locally depressed region (as defined by the parameter set) and exhibits a minor rebound. This design reduces reliance on static spacing under regime shifts.

Layer cap and sizing. The exchange DCA engine supports flexible layer caps (up to 50). In the present study, the automated configuration uses a fixed cap of $L_{\text{auto}} = 9$ layers. Accordingly, the boundary state $D = 9$ represents cycles that reach the cap (and, if any rare manual supplements occur, also aggregates $d > 9$ cycles). After this cap is reached, no further averaging is executed under the automated rule set. Supplementary manual entries, if any, are treated as boundary events and are not included in the automated execution logic analyzed here. Position size increases gradually with depth but deliberately avoids exponential doubling patterns associated with classical gambling martingales. Under the current spacing configuration (approximately 2–3% per layer), reaching the ninth layer corresponds to a cumulative downside displacement of roughly 15–25%. This window is intended to represent a moderate stress regime in high-volatility spot markets: large enough to exceed routine fluctuations yet sufficiently frequent to allow empirical observation. The associated percentage is configuration-dependent (spacing geometry and layer cap) and should not be interpreted as universal.

Exit logic. Exits are governed by an exchange-provided trailing take-profit mechanism. Once the weighted-average entry price reaches a predefined reward trigger, the trailing module activates and tracks the running maximum. A full exit occurs when price retraces by a specified offset. This integral exit rule enables additional reward capture during extended rebounds while preserving a consistent and auditable exit condition across cycles.

3.3. Execution cycle definition and depth encoding

An execution cycle is defined as the sequence from initiation of the first layer to the final take-profit exit. Let d_r denote the realized depth of the r -th completed cycle, defined as the number of activated layers prior to exit.

Depth is treated as a discrete structural state variable summarizing how far the execution process is pushed along an adverse price path within a cycle. For state-based analyses, we additionally define a truncated depth state

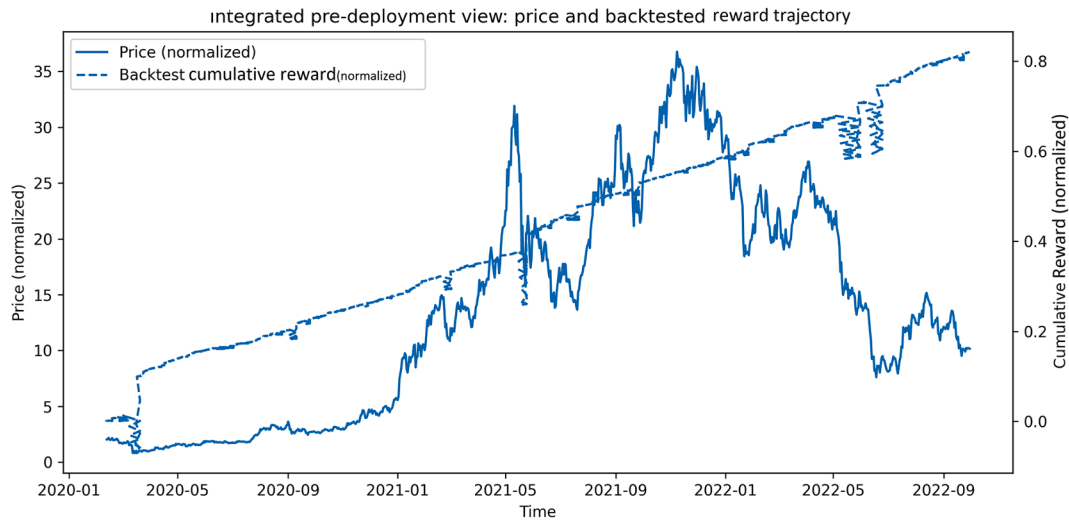


Fig. 1. Pre-deployment calibration context: market price and backtest-derived reward trajectory.

The figure shows the normalized daily price trajectory of the underlying asset (solid line, left axis) and the normalized cumulative reward trajectory obtained from TradingView backtesting under the same fixed entry/exit and position-sizing rules (dashed line, right axis) over the pre-deployment period (2020–September 30, 2023). The backtest is included only as a calibration reference based on information available up to the deployment date, and it is not intended as evidence of forward performance. Both series are normalized to preserve structural relationships while anonymizing absolute levels, providing the temporal and geometric context used to screen the configuration prior to live operation.

$$D_r = \min(d_r, 9),$$

so that $D_r = 9$ aggregates boundary-regime cycles with $d_r \geq 9$.

Empirically, most cycles resolve at shallow depths (typically $d_r \in \{1, 2, 3\}$), producing step-like increases in $\text{Reward}(t)$ that align with a stable linear envelope. Deep cycles occur infrequently but contribute larger rebound-related increments because recovery is realized from lower average entry levels. This encoding provides the basis for regime classification and transition analysis.

3.4. Regime encoding via a binary stress indicator

For analytical tractability, each completed cycle is mapped to a binary regime indicator representing whether the execution process entered a stress depth state. This transformation is specified ex ante as part of the experimental design and does not depend on realized reward magnitude or post hoc performance evaluation.

Specifically, cycles reaching or exceeding a predefined depth threshold are classified as stress-regime cycles, while all remaining cycles are assigned to a baseline regime. This binary variable encodes an internal structural condition (depth-state elevation) rather than a profitability outcome, enabling separation of routine volatility absorption from structurally stressed execution states.

3.5. Markov transition structure of cycle depth

(Index convention: r indexes completed cycles; $i, j \in \mathcal{S}$ index depth states.)

Let $D_r \in \mathcal{S}$ denote the truncated depth state of cycle r , with state space $\mathcal{S} = \{1, \dots, 8, 9\}$, where state 9 aggregates boundary-regime cycles. Under a fixed execution rule, each realized price-path segments maps to an observed depth state, producing a sequence $\{D_r\}_{r=1}^{N_{\text{cyc}}}$.

We summarize state dependence in the depth sequence using a first-order Markov transition matrix estimated by empirical frequencies:

$$T_{ij} = \Pr(D_{r+1} = j | D_r = i) = \frac{N_{ij}}{\sum_{l \in \mathcal{S}} N_{il}},$$

where

$$N_{ij} = \sum_{r=1}^{N_{\text{cyc}}-1} 1\{D_r = i, D_{r+1} = j\}.$$

This Markov representation is adopted as a compact descriptive summary of cycle-to-cycle depth dependence, not as a generative model of price dynamics [7,8].

3.6. Construction of fan-shaped depth-aligned trajectories

To visualize depth-dependent price deformation across heterogeneous cycles, we construct depth-aligned fan-shaped trajectories by normalizing price within each cycle relative to the initial entry price. The first layer is fixed at unity, and the horizontal axis represents discrete depth (layer index) rather than chronological time.

Aligning cycles by depth suppresses temporal heterogeneity and highlights cumulative structural burden induced by successive drawdowns. Each polyline corresponds to one cycle; shallow cycles may be subsampled for visual clarity. Extreme market events appear as trajectories that deviate from the normalized envelope formed by routine cycles, providing a structural bridge between execution mechanics (layer consumption) and observed stress patterns under extreme conditions.

4. Empirical results

This section evaluates the execution-level framework using live spot-

trading logs. The dataset spans 800+ days of continuous operation and includes $N_{\text{cyc}} = 305$ completed execution cycles for the representative Asset-A/USDT setting. Results are organized to match the theoretical components in Section 2 and the experimental encodings in Section 3.

Fig. 1 provides context for the survivability-filter calibration environment used prior to the analyzed live period. It is included to document system initialization (entry/spacing/trigger screening) rather than as a performance benchmark.

4.1. Depth distribution and operating-range concentration

Each completed execution cycle is characterized by its realized depth d_r , defined as the number of activated layers prior to exit (Section 3.3). Fig. 2 reports the empirical distribution of d_r over 305 cycles. The distribution is strongly right-skewed: more than half of the cycles terminate at depth 1, and approximately 82% resolve within depths 1–3. Deeper excursions occur progressively less frequently, with only a small fraction reaching the boundary regime ($d_r \geq 9$).

This concentration of mass in shallow depths is the primary empirical reason why cycle-aggregated reward accumulation can appear stable over long horizons: most execution activity is repeatedly resolved within a narrow, shallow operating range, whereas deep excursions appear as rare boundary events under stress.

4.2. Quadratic expansion of exposure with depth

For each cycle, executed layer sizes and cumulative deployed capital were recorded as a function of realized depth. Let $E(K)$ denote cumulative exposure after activating K layers (Section 2.2). Empirically, exposure growth is well summarized by a quadratic form,

$$E(K) \approx c_2 K^2 + c_1 K + c_0.$$

Fig. 3 plots exposure versus depth and shows a smooth convex profile consistent with this quadratic structure. Importantly, the growth is bounded by a fixed layer cap and is sub-exponential, distinguishing the configuration from doubling-type martingale schemes. In the representative dataset, quadratic fits achieve high goodness-of-fit, indicating that quadratic exposure growth emerges endogenously from the system's sizing/activation structure under the fixed configuration.

4.3. Depth-aligned fan geometry and the depth barrier

To examine how price-path deformation changes with depth, we construct depth-aligned fan trajectories by normalizing each cycle's price path to its cycle-start price (Section 3.6). Fig. 4 shows that cycles within shallow-to-intermediate depths form a coherent envelope: for depths up to approximately 7–8, normalized trajectories remain comparatively concentrated, consistent with routine volatility absorption within the designed operating range.

A qualitative transition appears near the boundary regime. At depth 9 (the aggregated $d_r \geq 9$ state), dispersion increases sharply and the fan envelope contracts, indicating that structurally favorable rebound space becomes constrained in depth-aligned coordinates. This pattern supports treating depth ≥ 9 as a boundary-state regime in the execution state space, rather than as a smooth continuation of shallow-cycle geometry.

Fig. 5 conditions outcome characteristics on realized depth. Shallow cycles are typically resolved quickly, whereas adverse cycle characteristics (e.g., extended time-under-stress) concentrate disproportionately near the boundary regime, consistent with the collapse of rebound space in the fan representation.

4.4. Markovian transition behavior of depth states

Because cycle depth forms a discrete state sequence across completed

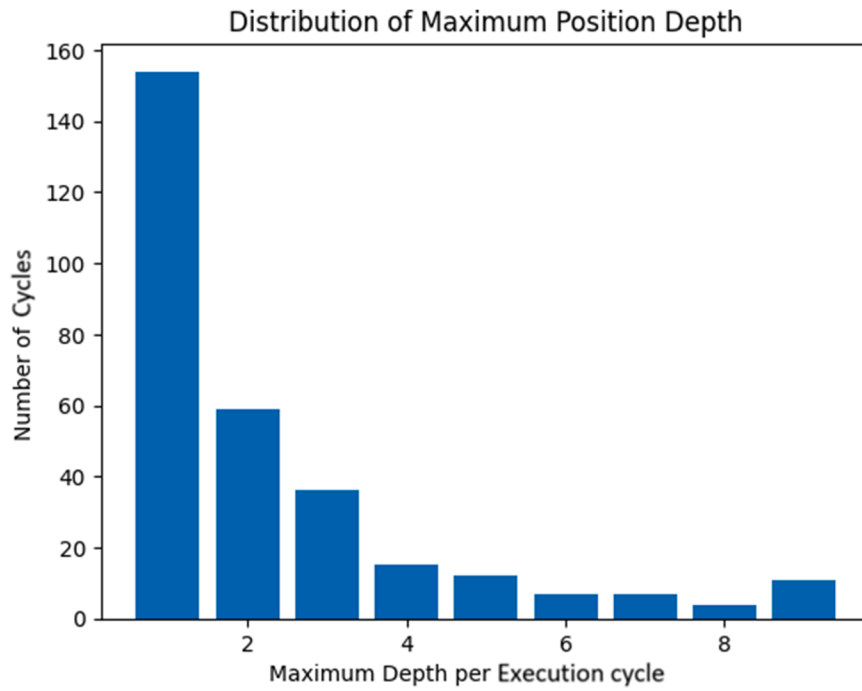


Fig. 2. Distribution of maximum position depth across execution cycles.

Each bar represents the number of execution cycles reaching a given maximum depth before termination. Depth values greater than or equal to nine are grouped as depth = 9, reflecting an operational boundary regime beyond the intended automated operating range. The distribution indicates that most cycles resolve at shallow depths, while deep-layer escalation occurs infrequently.

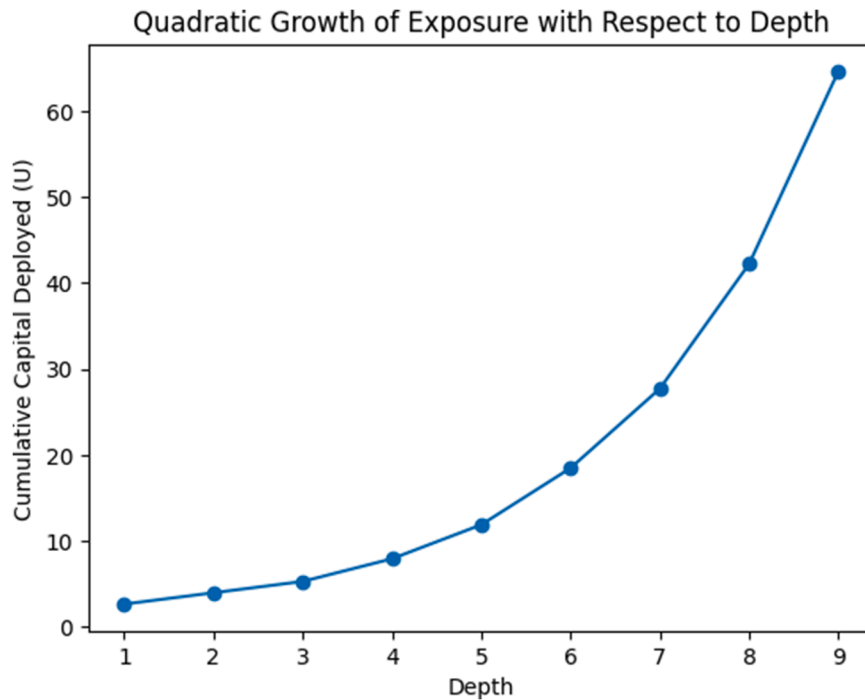


Fig. 3. Quadratic growth of cumulative exposure with respect to position depth.

The figure illustrates the cumulative capital deployed as position depth increases within a representative execution cycle. Exposure increases monotonically and exhibits a convex, approximately quadratic structure rather than exponential doubling, reflecting controlled capital escalation with an explicit depth limit.

cycles, we estimate a first-order Markov transition matrix on $S = \{1, \dots, 8, 9\}$ using the empirical frequency estimator in Section 3.5 [7,8]. Fig. 6 shows three robust features.

- (i) Shallow-state persistence: low-depth states exhibit high self-transition probability and high probability of returning to shallow states.
- (ii) Incremental movement within the interior: transitions among states 1–8 are predominantly local (adjacent or near-adjacent),

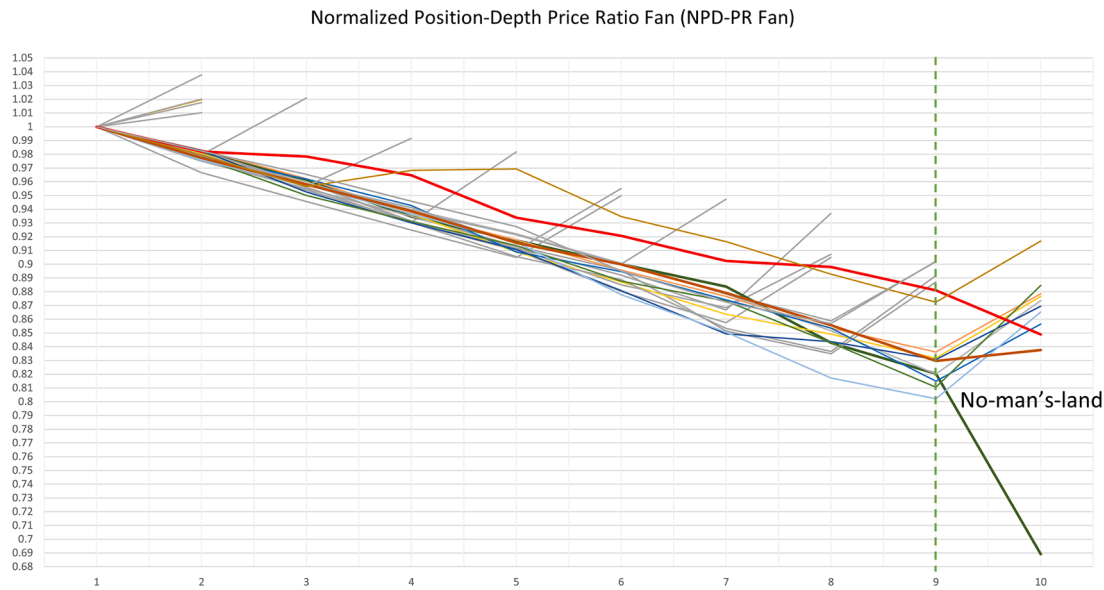


Fig. 4. Normalized price–depth fan highlighting the boundary-state regime. Prices at each depth are normalized by the first entry price of the corresponding execution cycle. Intermediate depths (1–8) exhibit highly similar geometric patterns and are therefore shown in muted tones. Cycles reaching depth 9 are emphasized, marking a boundary-state regime in which structurally favorable rebound space collapses. The dashed line indicates the depth boundary beyond which fully automated operation is not considered structurally stable.

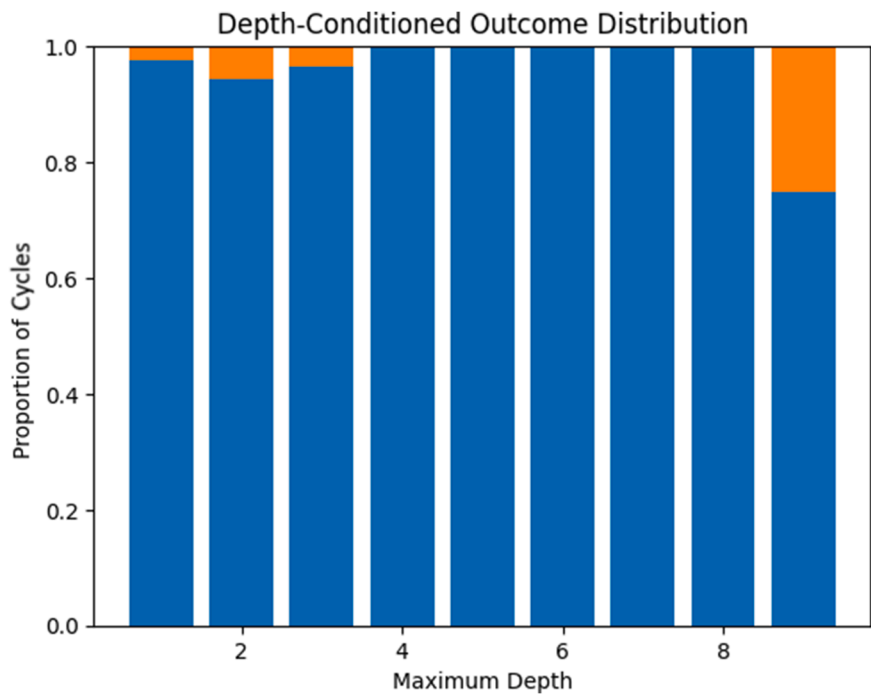


Fig. 5. Depth-conditioned outcome distribution across execution cycles. Execution cycles are grouped by their maximum depth, and outcomes are classified by whether the cycle resolved within 8 hours. Shallow-depth cycles exhibit highly stable outcomes, while adverse outcomes concentrate sharply near depth 9. This outcome degradation aligns with the boundary-state behavior identified in the normalized price–depth fan (Fig. 4).

- consistent with gradual variation in depth under ordinary conditions.
- (iii) Shock-driven boundary entry and post-resolution relaxation: transitions into the boundary state (9) are rare and often occur without a prolonged stepwise climb through intermediate depths, while exits from state 9 show elevated probability of reverting toward lower-depth states after resolution.

Overall, the transition structure supports interpreting depth as an execution-level structural state variable with regime dependence and inertia, rather than as an independent sequence of i.i.d. outcomes.

5. Discussion

The results in Section 4 support the interpretation that the behavior, risk, and survivability of layered-averaging systems are primarily

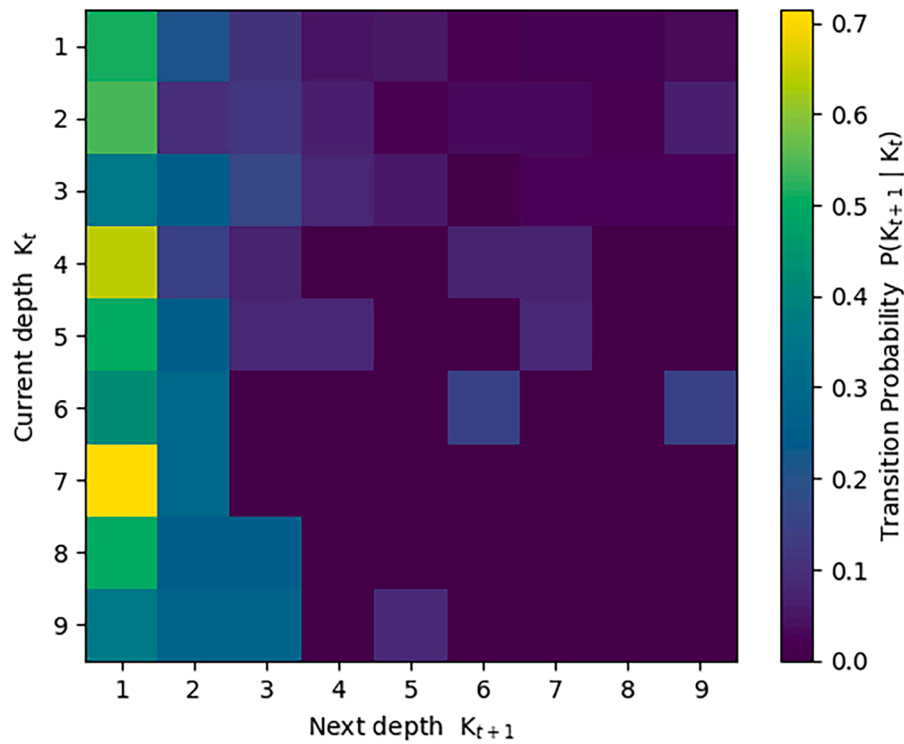


Fig. 6. Markov transition heatmap of depth states.

Each cell represents the conditional probability of transitioning from depth state D_t to D_{t+1} across consecutive completed execution cycles, i.e., $T_{ij} = \Pr(D_{t+1} = j | D_t = i)$. Probability mass concentrates in shallow-depth states, indicating structural stability under typical conditions. Transitions into deep-layer states are infrequent and are not preceded by gradual escalation, supporting a shock-driven rather than incremental interpretation.

governed by structural state evolution— i.e., how the system moves across depth states under bounded execution—rather than by probabilistic price prediction or outcome-frequency arguments. Here we synthesize the depth distribution, depth-aligned fan geometry, outcome asymmetry across depth regimes, and Markovian transition behavior into a unified execution-level interpretation. Collectively, these results motivate viewing layered-averaging execution as a state-constrained, curvature-sensitive system whose stability depends on bounded exposure growth, temporal flexibility, and structural alignment with draw-down–rebound segments.

5.1. Interpretation of depth ≥ 9 as a boundary-state regime

Across all empirical views, a critical threshold emerges around depth $d \geq 9$, beyond which system behavior changes qualitatively. This threshold is not merely “larger drawdowns” on a continuous scale; it corresponds to a regime shift in the internal depth-state dynamics of the execution process. Accordingly, we treat depth ≥ 9 as an operational controllability boundary (the “no-man’s-land” regime), conceptually distinct from statistical risk-boundary forecasts that estimate next-day upper/lower move limits under a chosen significance level [9]. Evidence for this boundary is consistent across three complementary views: (i) the depth distribution and depth-conditioned outcome spectra show that $d \geq 9$ cycles are rare and disproportionately adverse; (ii) the normalized price–depth fan exhibits envelope collapse beyond this region, indicating contraction of structurally favorable rebound space; and (iii) Markovian depth transitions suggest that entries into the boundary regime are typically shock-driven rather than incrementally accumulated, with exits often occurring via rapid resets after capitulation.

Deep-layer instability may also coincide with heightened market connectedness, where dependence structures deteriorate and rebound space compresses under systemic spillovers. Prior evidence that hedging/safe-haven properties in crypto-related markets can weaken

under stress is consistent with this interpretation [2]. In this sense, envelope collapse beyond the depth barrier is compatible with a systemic stress state in which deep-state persistence becomes quasi-absorbing from an operational perspective.

5.2. Structural (not probabilistic) meaning of layered-averaging dynamics

A defining element of the present analysis is cycle-internal normalization: within each execution cycle, the first entry price is normalized to unity and subsequent entries are expressed as relative price ratios. This removes absolute price levels and compresses heterogeneous regimes into a common execution coordinate system, isolating deformation patterns across depth states. Under this representation, layered averaging (often labeled “martingale” in trading practice, as defined in Section 1) is not interpreted through a price-prediction or outcome-frequency lens. Instead, it is treated as a geometric process evolving within a constrained state space: depth functions as a discrete state index; relative price deformation reflects curvature encountered as drawdowns are absorbed; and fan thickness represents the width of feasible structural responses. The fan geometry further indicates that vulnerability is not smoothly distributed across depth, but concentrates sharply as curvature steepens—manifesting as compression and envelope collapse—rather than as a gradual deterioration explainable by frequency arguments alone.

5.3. Layered averaging as practice-based “martingale,” but not doubling

Layered averaging in practice is often grouped under the colloquial label “martingale” because it accumulates exposure monotonically during adverse price movement. While the mechanism studied here overlaps with what many practitioners call “martingale-style” execution, it is explicitly not a classical doubling scheme (e.g., 1–2–4–8–16), which relies on exponential escalation as the primary recovery

mechanism [10,11]. Empirically, the capital allocation patterns examined here increase monotonically but sub-exponentially and are well-approximated by controlled quadratic accumulation under bounded execution constraints. This distinction is operationally central: doubling implicitly treats recovery as “forced” by geometric size escalation, whereas the present framework emphasizes execution geometry—deploying incremental exposure selectively when the drawdown–rebound path provides structurally usable rebound space, while avoiding prolonged persistence in deep states. Consistent with this, most cycles resolve at shallow depths, implying that viability is less attributable to any assumed inevitability of rebound than to (i) the empirical dominance of shallow resolutions and (ii) disciplined control of exposure growth and deep-state persistence.

5.4. Drawdown–rebound geometry and divergence points

Representative execution cycles (e.g., moderate-depth and deep-depth cycles) exhibit a recurring structural pattern. Price trajectories often begin with gradual downward displacement that can resemble intermediate “support,” but as curvature matures, displacement accelerates and pushes the system into deep drawdown zones where operational stress intensifies. Rebounds from these zones are frequently asymmetric and incomplete, resembling damped recovery rather than full reversal.

Layered-averaging execution exploits transient rebound windows by increasing exposure density nearer local minima, lowering the weighted-average entry price and enabling exits within limited rebound space. These exit zones should not be read as equilibrium points; rather, they are structural divergence points where the system’s execution geometry temporarily aligns with the local recovery curvature of the price trajectory. In practice, spacing, initiation, and exit constraints are tuned to increase the probability of engaging near such divergence points through long-horizon observation, retrospective analysis, and multi-parameter screening—a structural calibration process rather than a predictive one.

5.5. Markovian state transitions and system stability

Although market prices need not satisfy Markov assumptions, the depth evolution of the execution system exhibits structured, regime-dependent transition behavior. Modeling depth as a first-order conditional transition process in the depth-state space provides a compact, interpretable summary of internal state dependence. Two empirical features are especially salient: transitions from shallow to deep states are rare and often shock-driven, and post-shock resolutions frequently exhibit rapid contraction back toward shallow depths, consistent with normalization of volatility and liquidity conditions after extreme events.

This perspective clarifies why stability controls—such as entry spacing, cooldown intervals between cycles, and hard depth caps—are not heuristic add-ons but engineering responses to state-transition risk. In spot (non-leveraged) operation, the absence of liquidation introduces temporal degrees of freedom, enabling time-based mechanisms (e.g., suspension windows or delayed exits) to absorb structural stress when deep states are reached.

5.6. System stability and the role of structural cost

The representation

$$\text{Reward}(t) = at + b - R(t)$$

summarizes the empirical observation that $\text{Reward}(t)$ can evolve along a near-linear envelope under bounded execution, while deviations emerge under adverse regimes. The term $at + b$ captures baseline accumulation associated with repeated drawdown–rebound realizations under fixed exit constraints, whereas $R(t)$ aggregates execution-level

structural costs that become salient under stress, including extended time-under-stress, depth persistence, rebound insufficiency, and parameter–market misalignment. Thus, visually smooth $\text{Reward}(t)$ trajectories should not be interpreted as frictionless execution; near-linear accumulation is observed only after accounting for these state-dependent costs.

Finally, we do not claim that RSI-segmented trajectories obey a global quadratic law. The quadratic form is adopted as a minimal curvature proxy capturing the dominant convexity observed in depth-dependent exposure growth and drawdown–rebound segments. Within this framing, stability is associated with controllable depth evolution, bounded exposure growth, and sufficient temporal flexibility for recovery-based exits.

5.7. Risk sensing and adaptive spacing as controllability-preserving controls

Within the execution-level framing adopted here, risk is not positioned as a return-forecast target, but as a state-dependent operating condition that must be monitored to preserve controllability. The deviation term $R(t)$ aggregates log-constructible structural costs that emerge under stress. A practical implication is the need for an auxiliary risk-sensing layer that triggers conservative control adjustments before the system enters boundary-state regimes (Section 5.1). Such signals are most effective when auditable and directly actionable (e.g., widening spacing or temporarily pausing engagement), rather than opaque model recommendations, consistent with the value of interpretable mechanisms in high-stakes settings [12].

Risk sensing can be implemented using multiple indicators, and we do not claim a universal choice. Practitioners may use qualitative microstructure cues (e.g., accelerated downside displacement) or statistical tail-risk estimators as auxiliary warnings. For example, rolling-window downside-risk estimation has been proposed as a practical risk “traffic-light” for exposure adjustment [13], and related formulations estimate risk-boundary limits under chosen significance levels [9]. In the present framework, such external signals are treated strictly as auxiliary warnings: as stress escalates, widening spacing—and when necessary pausing execution—reduces deep-state persistence risk and slows the growth of $R(t)$. In this sense, spacing is an engineering control variable that maps stress escalation into conservative execution geometry and helps preserve rule-governed operation under adverse regimes.

6. Limitations and boundary conditions

Despite the structural regularities documented in this study, the proposed interpretation is intended to apply only within a bounded operational envelope. The purpose of this section is not to weaken the reported patterns, but to specify when the notion of controllability is meaningful and when it can break down.

First, the analysis is conditional on the absence of terminal exogenous failures within the observation horizon. Events such as exchange insolvency, settlement disruption, stablecoin/peg failure, or abrupt market-wide dislocation are largely external to the execution dynamics studied here and cannot be neutralized by internal rule design alone.

Second, the reported quadratic exposure geometry and near-linear $\text{Reward}(t)$ behavior are specific to spot, unleveraged execution. Introducing leverage changes liquidation mechanics, feasible control actions, and time-to-failure properties; therefore, the empirical relationships described here should not be extrapolated to highly leveraged derivatives without redesign and separate validation.

Third, a practical controllability boundary emerges in deep-layer regimes. Cycles approaching or exceeding approximately the ninth layer exhibit qualitatively different price–depth behavior (abrupt dispersion and reduced rebound space), under which continued rule-based accumulation can introduce disproportionate operational risk. We therefore interpret $\text{depth} \geq 9$ as an ex post, execution-log-defined

controllability limit rather than an ex ante forecasted threshold; its effective location may shift across venues, liquidity conditions, and microstructure regimes.

More broadly, extreme market dislocations can overwhelm any fixed boundary or learned regularity. Established frameworks—drawdown-based risk measures and sequential modeling approaches—advance downside-risk quantification and temporal dependence modeling, but do not remove vulnerability to discontinuous jumps, liquidity evaporation, or regime-dependent correlation spikes [3,14]. Accordingly, the boundary described here should be read as an operational limit under bounded conditions, not as a survivability guarantee under all crisis types.

Fourth, crossing the boundary does not imply “strategy failure” per se; it marks an intentional design limit where fully automated control is no longer structurally justified. Beyond this point, actions such as continued accumulation, exposure reduction, or temporary suspension become operator-dependent and are constrained by finite capital, liquidity, and heterogeneous risk tolerance; formal stop-loss and discretionary policies are therefore outside the scope of the present study.

Fifth, capital allocation is an external boundary condition. When the fraction of total capital allocated to the layered-averaging module is small, adverse excursions may be absorbed without systemic impairment; as allocation approaches saturation, buffering capacity diminishes and $R(t)$ can rise sharply under stress. No exposure geometry can compensate for capital saturation.

Sixth, exploratory tests with fixed stop-loss constraints exhibited instability when forced termination occurred during high-pressure drawdown phases. In spot settings, deep drawdowns do not necessarily imply immediate structural failure, but unresolved deep states can degenerate into passive inventory holding rather than active reward-generation cycles, requiring oversight beyond the automated execution dynamics analyzed here.

Finally, detailed executable parameterization is intentionally withheld for practical and ethical reasons: inappropriate parameter selection or capital misallocation in nonlinear, risk-sensitive environments can lead to severe losses. Consistent with high-stakes design principles, we emphasize interpretable, rule-governed control variables (spacing, depth caps, suspension) and log-computable proxies as a safety interface, rather than opaque decision pipelines [12].

Overall, the contribution is measurement-oriented rather than predictive: it characterizes how recurrent drawdown–rebound segments can yield near-linear $Reward(t)$ under bounded spot execution, and clarifies where and why these regularities degrade.

7. Conclusion

This study provides an execution-level, measurement-oriented account of why layered-averaging systems can exhibit stable, approximately linear $Reward(t)$ when operated under bounded constraints in nonlinear and volatile digital-asset spot markets. Using more than 800 days of continuous real-capital operation and 305 completed execution cycles, we document three empirical regularities:

- (i) drawdown–rebound price paths display pronounced nonlinear curvature;
- (ii) exposure accumulated through depth-dependent entries exhibits robust convex growth that is well approximated by a quadratic profile
- (iii) $Reward(t)$ concentrates around a stable near-linear envelope across cycles, including shock-driven drawdowns

These findings suggest that reward regularity is not primarily

attributable to forecasting accuracy. Rather, it emerges from repeated rule-governed entry and exit transformations applied to drawdown–rebound segments within a controllable depth range. To characterize departures from the idealized linear envelope, we introduce the deviation term $R(t)$, operationalized from execution logs as an empirical proxy for accumulated structural stress (e.g., time-under-stress, depth persistence, rebound insufficiency, and parameter–market misalignment). Depth-state transition analysis further indicates that extreme-depth cycles are rare and typically shock-driven, supporting the practical role of a deep-state boundary (e.g., $depth \geq 9$) as a limit of fully automated control.

Overall, the contribution is not prescriptive. Instead, it reframes layered averaging as a state-dependent stability problem: sustainable operation depends on bounded exposure growth, temporal flexibility, and disciplined state management under adverse regimes. By making structural costs explicit through $R(t)$ and identifying a controllability boundary for extreme-depth states, the study clarifies how near-linear $Reward(t)$ can arise in practice from nonlinear market dynamics under bounded execution constraints.

7.1. Future research

Several extensions merit further investigation. First, the deep-state boundary could be characterized more formally in relation to risk-boundary estimation and dependence-breakdown perspectives. Second, multi-asset extensions of the depth-state framework—together with connectedness-aware stress modeling—may clarify how systemic spillovers interact with boundary-state persistence. Third, alternative empirical constructions of $R(t)$ (e.g., hazard-style or Bayesian updating formulations) could better link execution-state behavior to macro-regime conditions. Finally, incorporating interpretable signal layers may help clarify the trade-off between rule-governed intervention and predictive modeling in nonlinear markets.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the author used ChatGPT (OpenAI) and Claude (Anthropic) to assist with language refinement, structural organization, and improvement of academic clarity. These tools were used exclusively to support the presentation of ideas, not for data generation, data analysis, or result interpretation. After using this tool, the author carefully reviewed, edited, and validated all content, and takes full responsibility for the accuracy, originality, and integrity of the published work.

Funding acquisition

None.

CRedit authorship contribution statement

Li-Yung Chen: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Log-constructible proxy family for reward deviation $R(t)$ (diagnostic form)

A.1. Purpose and scope

This appendix provides an operational proxy family to instantiate the reward deviation term $R(t)$ introduced in Section 2.4. The intent is diagnostic and audit-oriented: to ground “execution-level structural stress” in observable quantities that can be computed directly from execution logs. The proxies are not proposed as universal downside-risk metrics (e.g., VaR/ES/EVaR) and are not intended as forecasting signals. No component weights are calibrated in this study; the purpose is interpretability and traceability rather than optimization.

A.2. Notation (cycle-level quantities)

Let i index completed execution cycles. Let t_i^{start} and t_i^{exit} denote cycle start and exit times. Let $P_{0,i}$ be the cycle-start reference price, $P_{\min,i}$ the minimum traded price before exit, and $P_{\text{exit},i}$ the realized exit price under the predefined take-profit rule. Let L denote the configured layer limit and $K_i \leq L$ the realized number of filled layers in cycle i . Let $1\{\cdot\}$ denote the indicator function.

A.3. Boundary-condition parameter (capital allocation ratio)

(BC1) Capital allocation ratio (operating envelope)

Let B_{total} denote total available capital under operational constraints and B_{LA} the maximum capital allocated to layered execution. Define

$$\rho = \frac{B_{\text{LA}}}{B_{\text{total}}}.$$

ρ is treated as a boundary-condition parameter that governs the feasible operating region (e.g., depth tolerance, likelihood of extended holding, and exposure to excess-depth cycles). In the present study, ρ is used to describe the operating envelope and does not enter the deviation accumulator additively.

A.4. Core proxy components (log-constructible stress realizations)

(A1) Excess-depth beyond designed coverage (spacing tightness / coverage miss)

Define realized downside magnitude

$$D_{\text{mkt},i} = 1 - \frac{P_{\min,i}}{P_{0,i}}.$$

Given per-layer spacing x and layer limit L , the designed coverage depth is

$$D_{\text{max}} = 1 - (1 - x)^{L-1}.$$

Define excess depth

$$X_i = \max \{0, D_{\text{mkt},i} - D_{\text{max}}\}.$$

Interpretation: $X_i > 0$ indicates the realized downside exceeded the designed layer coverage, typically manifesting as reward-trajectory flattening or delayed recovery in the corresponding segment.

(A2) Exposure concentration in deep layers (deep-state loading)

Let u_k denote the position size at layer k . Choose a deep-layer threshold k^* (e.g., deepest 30% of layers). Define

$$C_i = \frac{\sum_{k=k^*}^L u_k}{\sum_{k=1}^L u_k}.$$

Interpretation: larger C_i indicates heavier concentration in deep layers, which increases sensitivity to depth saturation and rebound insufficiency near the controllability boundary.

(A3) Time-under-stress (prolonged holding beyond reference)

Define holding time

$$H_i = t_i^{\text{exit}} - t_i^{\text{start}}.$$

Given a reference duration H_0 (e.g., median holding time under stable regimes), define

$$U_i = \max \{0, H_i - H_0\}.$$

Interpretation: larger U_i corresponds to prolonged cycles, capturing persistence of adverse regimes and delayed exit under fixed take-profit constraints.

(A4) Entry clustering (insufficient cooldown / accelerated layer consumption)

Let $\Delta t_{i,k}$ denote the time gap between consecutive entries in cycle i . Given a cooldown threshold τ , define

$$F_i = \frac{1}{K_i - 1} \sum_{k=1}^{K_i-1} 1\{\Delta t_{i,k} < \tau\}.$$

Interpretation: larger F_i reflects clustered entries (rapid consumption of layers), which increases the likelihood of approaching depth limits within a single shock phase.

(A5) Inter-cycle overlap (insufficient recovery between cycles)

Define the separation between consecutive cycles

$$\Delta\tau_i = t_{i+1}^{\text{start}} - t_i^{\text{exit}}.$$

Given a minimum separation threshold τ_{sep} , define

$$\mathbf{M}_i = 1\{\Delta\tau_i < \tau_{\text{sep}}\}.$$

Interpretation: frequent overlap compounds exposure and operational load across cycles, increasing aggregate deviation even if individual cycles remain rule-governed.

(A6) Rebound deficit (insufficient rebound amplitude delaying exit)

Define a rebound amplitude measure $\mathbf{A}_i$ computed from logs after the cycle minimum. A practical operational choice is a percentage rebound from the minimum to the realized exit:

$$\mathbf{A}_i = \frac{P_{\text{exit},i} - P_{\min,i}}{P_{\min,i}}.$$

Given a reference rebound scale $\mathbf{A}_0$, define

$$\mathbf{V}_i = \max\{0, \mathbf{A}_0 - \mathbf{A}_i\}.$$

Interpretation: larger $\mathbf{V}_i$ indicates rebound insufficiency relative to the exit requirement, typically associated with extended holding and delayed reward realization.

A.5. Optional extensions (kept as a diagnostic menu, not required in main analysis)

Depending on data availability and venue preference, additional log-derived diagnostics may be included without altering the conceptual role of $\mathbf{R}(t)$. Examples include:

- Slippage / spread stress: realized execution price deviation from intended limit levels during shocks.
- Order-fill failure / partial-fill intensity: fraction of layers with partial fills or delayed fills beyond a threshold.
- Funding/fee drag proxies (if applicable): cumulative fees during prolonged holds (spot fees, borrowing/funding if any).
- Volatility-state mismatch: realized volatility during cycle relative to configuration-implied tolerance bands.

These extensions are not used to claim completeness; they are presented as auditable indicators that may help explain envelope deviations under different operational settings.

A.6. Aggregation (uncalibrated, interpretable accumulator)

A minimal uncalibrated form is

$$\mathbf{R}(t) = \sum_{i:t_i \leq t} (\mathbf{X}_i + \mathbf{C}_i + \mathbf{U}_i + \mathbf{F}_i + \mathbf{M}_i + \mathbf{V}_i + \dots),$$

where “...” denotes optional components as in Section A.5. Component weights are not calibrated in this study. The intended use is interpretability: elevated $\mathbf{R}(t)$ explains why segments of the reward trajectory flatten or bend downward relative to the near-linear envelope. When stress components diminish, deviation accumulation slows and near-linear reward behavior re-emerges.

A.7. Implementation note (auditability)

All components above are computable directly from execution logs containing timestamps, filled layer prices, filled sizes, and exit times/prices. The design emphasis is transparency: each unit of deviation can be traced to specific realized operational events (excess-depth, prolonged holding, clustered entries, overlap, or rebound deficit), rather than inferred from model-based forecasts.

References

- [1] L. Kristoufek, What are the main drivers of the Bitcoin price? Evidence from wavelet coherence analysis, *PLoS. One* 10 (4) (2015) e0123923, <https://doi.org/10.1371/journal.pone.0123923>.
- [2] E. Bouri, S.J.H. Shahzad, D. Roubaud, L. Kristoufek, B. Lucey, Bitcoin, gold, and commodities as safe havens for stocks: new insight through wavelet analysis, *Q. Rev. Econ. Financ.* 77 (2020) 156–164, <https://doi.org/10.1016/j.qref.2020.03.004>.
- [3] A. Chekhlov, S. Uryasev, M. Zabarankin, Drawdown measure in portfolio optimization, *Int. J. Theor. Appl. Finance.* 8 (01) (2005) 13–58, <https://doi.org/10.1142/S0219024905002767>.
- [4] B. Mandelbrot, The variation of certain speculative prices, *J. Bus.* 36 (4) (1963) 394–419, <https://doi.org/10.1086/294632>.
- [5] R. Cont, Empirical properties of asset returns: stylized facts and statistical issues, *Quant. Finance* 1 (2) (2001) 223–236, <https://doi.org/10.1088/1469-7688/1/2/304>.
- [6] J.-P. Bouchaud, M. Potters, *Theory of Financial Risk and Derivative Pricing: From Statistical Physics to Risk Management*, 2nd ed., Cambridge University Press, Cambridge, UK, 2003.
- [7] J.D. Hamilton, A new approach to the economic analysis of nonstationary time series and the business cycle, *Econometrica* 57 (2) (1989) 357–384.
- [8] M. Kijima, *Markov Processes for Stochastic Modeling*, Chapman & Hall, London, UK, 1997.

- [9] I. Choi, W.C. Kim, Practical forecasting of risk boundaries for industrial metals and critical minerals via statistical machine learning techniques, *Int. Rev. Financ. Anal.* 94 (2024) 103252, <https://doi.org/10.1016/j.irfa.2024.103252>.
- [10] W. Feller, *An Introduction to Probability Theory and its Applications*, 3rd ed., 1, Wiley, New York, NY, USA, 1968.
- [11] E.O. Thorp, Optimal gambling systems for favorable games, *Rev. Inst. Int. Stat.* 37 (3) (1969) 273–293.
- [12] C. Rudin, Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead, *Nat. Mach. Intell.* 1 (5) (2019) 206–215, <https://doi.org/10.1038/s42256-019-0048-x>.
- [13] I. Choi, W.C. Kim, Estimating historical downside risks of global financial market indices via inflation rate-adjusted dependence graphs, *Res. Int. Bus. Finance* 66 (2023) 102077, <https://doi.org/10.1016/j.ribaf.2023.102077>.
- [14] T. Fischer, C. Krauss, Deep learning with long short-term memory networks for financial market predictions, *Eur. J. Oper. Res.* 270 (2) (2018) 654–669, <https://doi.org/10.1016/j.ejor.2017.11.054>.